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A methodological framework for a quantitative assessment of new technologies to boost the interoperability of railways services

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Abstract

This work aims to present a methodological framework to assess to what extent innovative technologies meet the needs of Travelers and TSPs involved in the digital ecosystem for door-to-door trips in Europe, making railways and public transport more attractive. Two kinds of quantitative data are used, operational KPIs and USI surveys. The Effectiveness, as a metric of the capacity to meet these needs by the innovative technology, is calculated by merging both data. The method considers travelers' socio-demographic profiles, allowing comparisons among TSPs and profiles for a specific technology. This work belongs to the methodological framework of the IP4MaaS project, which sets 6 demo sites on which this assessment method will be applied in a further stage. This work is interesting for TSPs and IT developers, researchers, policymakers, and organizations in the transport sector.

Keywords: User Satisfaction Index; Operational KPI; Effectiveness; Innovative technologies; Railways; MaaS.

Nomenclature

EU	European Union
IP4	Innovation Program 4
USI	User Satisfaction Index
KPI	Key Performance Factor
TSP	Transport Service Provider
GHG	Greenhouse Gases
LBE	Location-Based Experience
JP	Journey Planner
MaaS	Mobility as a Service

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1. Introduction

While cities increase their number of inhabitants, some rural areas suffer from depopulation. According to the United Nations, around 75% of Europe's population currently resides in urban areas, and a project is that by 2050 this figure will be almost 84% (United Nations. Department of Economic and Social Affairs, 2018). Moreover, the population is becoming older: In 2018, nearly 19.7% of the EU's population was aged 65 or over, representing a 2.6% increase since 2008 (EUROSTAT 2019)

Changes in the living environment and age of the population, and consequences in the travel patterns, will become the need for innovative solutions to fit the current needs better. Concerning the recent development and advancement of technologies, using clean public transport has become one of the main focuses of current environmental issues.

This work aims to present a methodological framework to assess, in a quantitative way, how innovative technologies can respond to the needs of travelers and Transport Service Providers (TSPs) involved in the digital ecosystem for door-to-door travels in Europe, making railways and public transport more attractive.

This work assesses the needs of travellers belonging to different socio-demographic profiles and TSPs (Transport Service Providers) based on rail transport in a methodological way addressing societal trends such as the reduction of Greenhouse Gases (GHG) emissions and road congestion (Bamberg et al. 2011).

The concept of the attractiveness of rail and public transport depends on complex psychological factors from a scientific and technical perspective (European Rail Research Advisory Council 2020); This assessment methodology consolidates the concept of "user profile" (COHESIVE project 2020), and the ability of the system to respond to the needs and expectations including socio demographic-related aspects such as aging, reduced mobility, impairments, and other specific conditions.

Terms as the Effectiveness of IT (Information Technology) solutions, the adaptability of data to the satisfying requirement of both travellers and operators, the equity of IT deployment in society (Misra and Panda 2017), and the potential acceptance by the market and, in particular, by railway operators and their ecosystem (Wolfram et al. 2016) are considered in the assessment methodology introduced by this work.

This assessment methodology belongs to the methodological framework developed in the first phase of the H2020 Shift2Rail project named IP4MaaS (www.ip4maas.eu), which sets 6 demo sites (Barcelona, Padua, Athens, Liberec, Osijek, and Warsaw) in which this methodology will be applied in a second stage.

This assessment framework has two strengths that make it unique in comparison with similar ones in the field:

- The combined computation of operational KPIs gathered from the performance of the IT solution while being used by users (travellers and operators), jointly with satisfaction indicators gathered through surveys filled by these same users; which provides a final quantitative metric about the benefit offered by this IT solution used in the public transport, especially by railways.
- The possibility of carrying out specific assessments per 'users' profiles, such as elderly, disabled people, or gender, based on their particular needs and expectations, to build equity by future improvements of the IT solution based on the results of this assessment.

2. Methodology

The methodology used in this paper sets the concept of "demonstration scenario" (IP4MaaS project. D3.1, 2021), which is the intersection of a new technology offered to travelers and a TSP (Transport Service Provider) that is offering it, and the "Effectiveness" as a metric on how this new technology is matching needs and expectations of these travelers and TSPs (Transport Service Provider), from the perspective of an aggregated analysis and per profiles of travelers in intersectional analysis. (Ahmadi and Hooshangi 2015).

To calculate the "Effectiveness" of new technology in a specific "demonstration scenario," two kinds of quantitative data, one objective and another subjective, are required (Ngossaha et al. 2017):

2.1. Objective data: Operational KPIs

A list of operational KPIs (Key Performance Indicators) measuring the gain of benefit for the TSP (Transport Service Provider) and the traveler by integrating innovative technology into the TSP's services (Litman 2016) can be

identified in the table 1. Innovative technologies listed in this table are new functionalities developed in Innovation Program number 4 (IP4) according to the Shift2Rail program (Shift2Rail Master Plan 2015). These operational KPIs will be automatically collected from the cloud system of these IP4 functionalities.

Table 1. List of operational KPIs per functionalities

Number	Innovative Technology (IP4)	KPI	Units
1	Journey Planner (JP)/ Offer Builder	Number of involved modes of transport in the trip (multimodality)	Number per day
2	Journey Planner (JP)/ Offer Builder	TSP Web-services acting as JP integrated into the IP4 ecosystem	Number of TSP integrated
3	Journey Planner (JP)/ Offer Builder	A successful proposal for customers (due to the integration of transport modes)	Number of travel solutions shown per day
4	Journey Planner (JP)/ Offer Builder	Available travel solutions for customers (due to the integration of transport modes)	Number of travel solutions shown per day
5	Booking	Number of offers booked per day	Number of offers booked per day
6	Issuing	TSP web-services for issuing process integrated into the IP4 ecosystem	Number of TSP integrated
7	Issuing	Successful issuing of multimodal travel solutions	Number of issues per day
8	Validations and Inspection	Total number of Ticket(s) purchased	Number of tickets validated per day
9	Trip Sharing	Number of trips shared by more than one traveler	Number of trips shared
10	Location-Based Experiences (LBE)	Number of users using the entertainment services	Number of users per day
11	Location-Based Experiences (LBE)	Time using the entertainment services	Number of seconds per connection
12	Trip Tracking	TSP locations (stations, platforms) available for navigation	Number of TSP locations
13	Trip Tracking	Successful delivery of notifications on the status of a planned trip	Number of successful notifications per day
14	Ancillary Services	TSP web-service integrated for shopping of ancillary services	Number of TSP integrated
15	Ancillary services	Shopping for ancillary services through the integration of a static list of available services	Number of items bought from the list per day
16	Alternatives calculation	Service offerings to travellers (in case of disruption)	Number of services per day

Each KPI is measured in a specific unit, and it will be converted into a dimensionless KPI, with a value between 0 and 1, by dividing each KPI linked to a specific functionality by the maximum value measured for this KPI among all TSPs. The KPI is defined in a mean that the higher, the better, so a dimensionless value close to 1 will always be better than a dimensionless value close to 0 (IP4MaaS project. D4.2, 2022).

Per each demonstration scenario "D" or living lab, several User Journeys "i" were considered with a different travel solution (Scheiner 2006) to go from an origin to a destination through the combination of several means of transport (TSP "k"). The previous list of KPIs was proposed based on the technological capabilities of the TSP "k"

for the integration of an IP4 functionality "j" (IP4MaaS project. D3.1. 2021), as depicted in Figure 1.

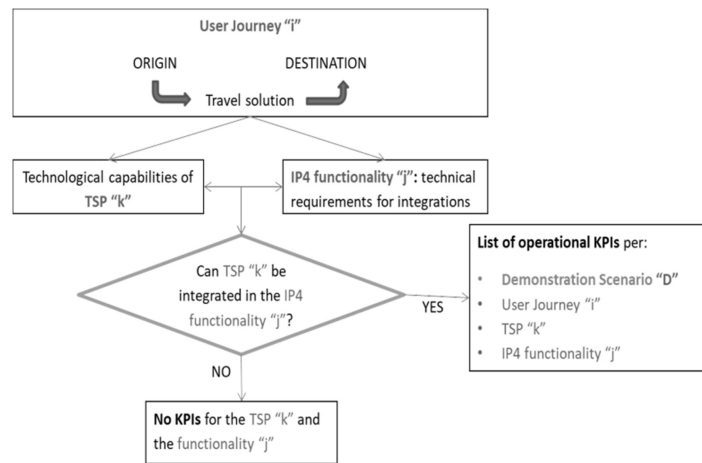


Figure 1. Flowchart to define KPIs

2.2. Subjective data: User Satisfaction Index (USI) surveys

Two User Satisfaction Indexes (USIs), one for travelers and another for TSPs, measure their opinions about a specific new technology's utility (Bamberg et al. 2011). These indexes are calculated from answers to surveys:

- The USI questionnaire for travelers evaluates questions about needs and expectations from 1 (strongly disagree) to 5 (strongly agree) and collects socio-demographic information of the traveler to conduct an intersectional analysis (Misra and Panda 2017) (Pelangi et al., 2021).
- The USI questionnaire for TSPs contains questions about the satisfaction of the TSP with the new integrated technology (Maranghi et al., 2020).

The USI questionnaire for travelers is structured in 2 sections:

- Part A: questions about needs and expectations. This questionnaire includes a specific section with questions for all profiles and a second section with profile-based questions, which will be only answered by users that select this profile in the socio-demographic questionnaire.
- Part B: Socio-demographic questionnaire for travelers that will support the analysis of satisfaction per each profile vector "r," as a set of socio-demographic characteristics of the traveler (IP4MaaS project. D3.1.2021).

The USI questionnaire for 'traveler's aims to evaluate their satisfaction level with those IP4 functionalities related to travelers among those listed in table 1 and others for which an operational KPI has not yet been defined (MaaSive Project 2020):

1. Journey planning function: The utility of finding routes involving different modes of transport (Ahmadi and Hooshangi 2015)
2. Booking function: The utility reserves and purchases both a specific ticket and multimodal tickets (Han et al. 2020)
3. Issuing function: The utility to provide online tickets that can be validated and inspected through the mobile application
4. Mobility package's function: The utility offers the packages of trips with better offers to go from an origin to a destination.
5. Validation and inspection function: The utility to validate and inspect digital tickets (QR code, Voucher) through the application.
6. Trip tracking function: The utility to observe real-time information about your trips, such as the expected time of arrival, expected time of departure, the timetable, and traffic information

7. Alternative's calculation function: The utility help finds alternative accommodation in case of a trip disruption.
8. Location-based experience (LBE) function: The utility to discover entertainment services, such as quiz games or commercial offers provided during the trip.
9. Navigation function: The utility to guide and direct travelers in using the correct metro or bus stop, especially in the interchanges among different means of transport
10. Traveler's feedback function: The utility that allows travelers to submit or provide feedback about delays, cleanness of stations, disruptions, and crowdedness on public transportation or road environment those travelers might consider helpful for other travelers.
11. Trip sharing function: The utility allows travelers to share their trip with other travelers or friends and family (car sharing, bike sharing, and taxi sharing).
12. Guest user function: The utility that allows travelers to use the application as a guest and the features and accessibility provided in the application when travelers use it as a guest.
13. Preferences and Profiles function. The utility allows travelers to observe and check the quality level of provided features on the travelers' Profile application.
14. Group traveling function: The utility allows travelers to organize group travels with other travelers.
15. LBE functionality (Location-based experience): The function allows for building Location-Based Experiences for the user.

While the USI questionnaire for TSPs aims to evaluate the satisfaction level with those IP4 functionalities related to TSPs among those listed in table 1 and others for which an operational KPI has not yet been defined (MaaSive Project 2020):

1. Asset manager: The function to insert and describe TSP (Transport Service Provider) services and facilities in the IP4 platform (IP4MaaS project 2021).
2. Contractual Management Market Place: The function describes products integrated with mobility packages and proposes new mobility packages to other stakeholders.
3. Business Analytics function: The function to perform a descriptive, diagnostic, predictive, and prescriptive analysis of the data; to visualize data through dashboards, portals, or virtual reality; and use algorithms to anonymize the provided data. (Ahmadi and Hooshangi, 2015).
4. CEP configuration: The function to define and modify the rules based on Trip tracks.
5. Validation and Inspection: The function to control fraud from Travelers during their journey.

One example question of USI for travelers Part A is shown in figure 2, while figure 3 shows two example questions of Part B. Figure 4 introduces one example question of USI for TSPs:

The *Journey planning function* to find routes involving different modes of transport in a journey from an origin to a destination.

1.a- is user friendly, and in general, I am satisfied with its outputs

1. Strongly disagree	2. Disagree	3. Neither agree nor disagree	4. Agree	5. Strongly agree
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1. b- will save more time and money for you

1. Strongly disagree	2. Disagree	3. Neither agree nor disagree	4. Agree	5. Strongly agree
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Figure 2.- One example question of Part A in USI for Travelers.

Do you consider yourself to live in: a. A rural environment b. An urban environment c. A suburban environment d. Abroad/tourist	Please choose your age group. a. 18-24 years b. 25-44 years c. 45-64 years d. 65 years or more e. Prefer not to answer
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Figure 3.- Some socio-demographic questions of Part B in USI for travelers.

The *Asset Manager* functionality to insert and describe your services and facilities:

a.- is user friendly, and in general, I am satisfied with its outputs

1. Strongly disagree	2. Disagree	3. Neither agree nor disagree	4. Agree	5. Strongly agree
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b.- will allow you to know better your services offer and technology level

1. Strongly disagree	2. Disagree	3. Neither agree nor disagree	4. Agree	5. Strongly agree
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Figure 4.- One example question of USI for TSPs.

The satisfaction index for travelers belonging to a profile vector "r" with the functionality "j" offered by the TSP "k" is calculated as:

$$USI_{Traveller_{rjk}} = \frac{\sum_{w=1}^{m_{rjk}} \sum_{v=1}^{n_{1jk}+n_{2jk}} Score_{question_{wv}}}{m_{rjk} \cdot (n_{1jk} + n_{2jk}) \cdot 5} \quad (\text{Eq. 1})$$

Being:

$Score_{question_{wv}}$ = the score to the question "v" by the respondent "w".

n_{1jk} = number of questions applicable to all the profiles measuring the satisfaction with the functionality "j" offered by the TSP "k".

n_{2j} = number of questions apply only to the profile "r" measuring the satisfaction with the functionality "j" offered by the TSP "k".

m_{rjk} = number of respondents to the USI questionnaire belonging to the profile "r" measuring the satisfaction with the functionality "j" offered by the TSP "k".

The satisfaction index for a TSP "k" regarding a functionality "j" is calculated as:

$$USI_{TSP_{jk}} = \frac{\sum_{v=1}^{n_j} Score_{question_v}}{n_j \cdot 5} \quad (\text{Eq. 2})$$

Being, $Score_{question_v}$ the score to the question number "v" and " n_j " The number of questions in the USI questionnaire belonging to a specific functionality "j" offered by the TSP "k".

In both equations, a 5 appears to divide to normalize and obtain a value between 0 and 1 because the answer to each question has a value between 1 (representing the minimum satisfaction) to 5 (representing the maximum satisfaction).

All this quantitative data (operational KPIs and USIs) is handled together within the concept of "Effectiveness".

2.3. The concept of Effectiveness

The *Effectiveness* of a functionality "j" offered by a TSP "k" for a specific profile "r" in a demonstration scenario "D" is calculated through the following equation:

$$Effectiveness_{rjk} = \frac{\sum_{n=1}^N KPI_{nTSP_{jk}} + USI_{Traveller_{rjk}} + USI_{TSP_{jk}}}{N+1+1} \quad (\text{Eq. 3})$$

Being:

"N" is the number of operational dimensionless KPIs linked to the functionality "j" offered by the TSP "k" (N can be zero for some functionalities),

$KPI_{nTSP_{jk}}$ the value of the KPI "n" belonging to the functionality "j" offered by the TSP "k",

$USI_{Traveler_{rjk}}$ the value calculated in Eq. 1, and
 $USI_{TSP_{jk}}$ the value calculated in Eq. 2.

Given that the *Effectiveness* is dimensionless with a value between 0 and 1, the higher, the better, and different demonstration scenarios "D" can be compared to analyze how the needs of travelers in different locations or demo sites are matched by the same innovative technology "j" offered by different TSPs.

The three elements in the nominator are summed in a linear way and with an equal weight because an innovative technology with no good operational KPIs, no good acceptance level by travelers, or no good acceptance level by the TSP would not be implemented in practice or would not remain in use for a long time, as it would therefore not be answering users' needs. However, this linear distribution of weights among KPIs and USIs in the *Effectiveness* formula could be a limitation of the methodology due to the different amounts of data available per each term, which perhaps will require some refinements in the future when this data is available and handled in the IP4MaaS project.

This *Effectiveness* will be used in practice as a quantitative metric on the benefit that an IP4 function brings to users, travelers and TSPs. Low values of this *Effectiveness*, when compared between IP4 functions, TSPs, or different traveler profiles, will show weaknesses in the degree to which these IP4 functions meet the needs and expectations of travelers and TSPs, which will allow IT developers to implement improvements based on these results and boost the interoperability of railways services by creating more equitable IP4 functions.

3. Conclusion and further developments

A methodological assessment framework to measure in a quantitative way how the extent to which some innovative technologies developed by the IP4 Shift2Rail program meet needs of travelers and TSPs is presented in this work

With this aim, a new concept to define demonstration scenarios on which the assessment is conducted was introduced, jointly with two quantitative types of data: Operational KPIs and USIs, which allow the calculation of the Effectiveness of a specific innovative technology offered by a TSP to a profile group of travelers.

The Effectiveness is dimensionless and has a value between 0 and 1; the higher, the better, which measures how an innovative technology matches the needs and expectations of its users, travelers, and TSPs. For a specific technology, it allows comparisons between demonstration scenarios or TSPs and different profiles of travelers.

Future works will apply this quantitative assessment methodology to 6 demo sites with multiple demonstration scenarios set by the H2020 Shift2Rail IP4MaaS project to validate its benefits, proceed with the required refinements, and explore its potential.

Moreover, statistical correlations between operational KPIs and USIs could be defined by applying Machine Learning techniques as Bayesian Networks.

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